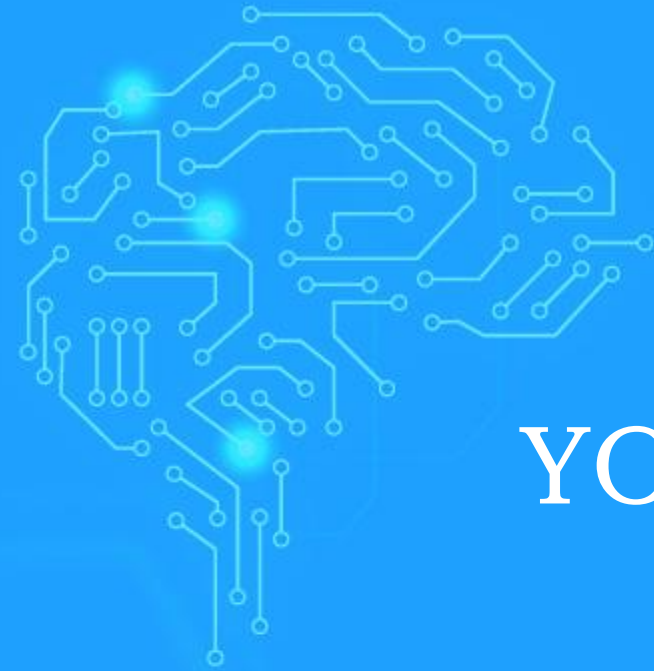




YOUR TECHNOLOGY & INNOVATION PARTNER

Tool Integration with MCP & Guardrails

Agenda



Recap

Model Context Protocol

Guardrails

Q&A

Recap - the five components

BRAIN



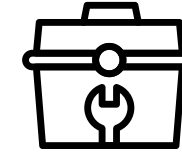
The brain is where an agent's **thinking** occurs, enabling it to process information and make informed decisions.

SENSES



Senses allow agents to **perceive** their environment by gathering data through various inputs, such as sensors and information feeds.

HANDS



The hands represent the agent's ability to **act** on decisions, executing tasks that were analyzed and planned.

GUARDRAILS



Guardrails ensure agents operate within **constraints**, providing boundaries that prevent harmful or undesirable actions from occurring.

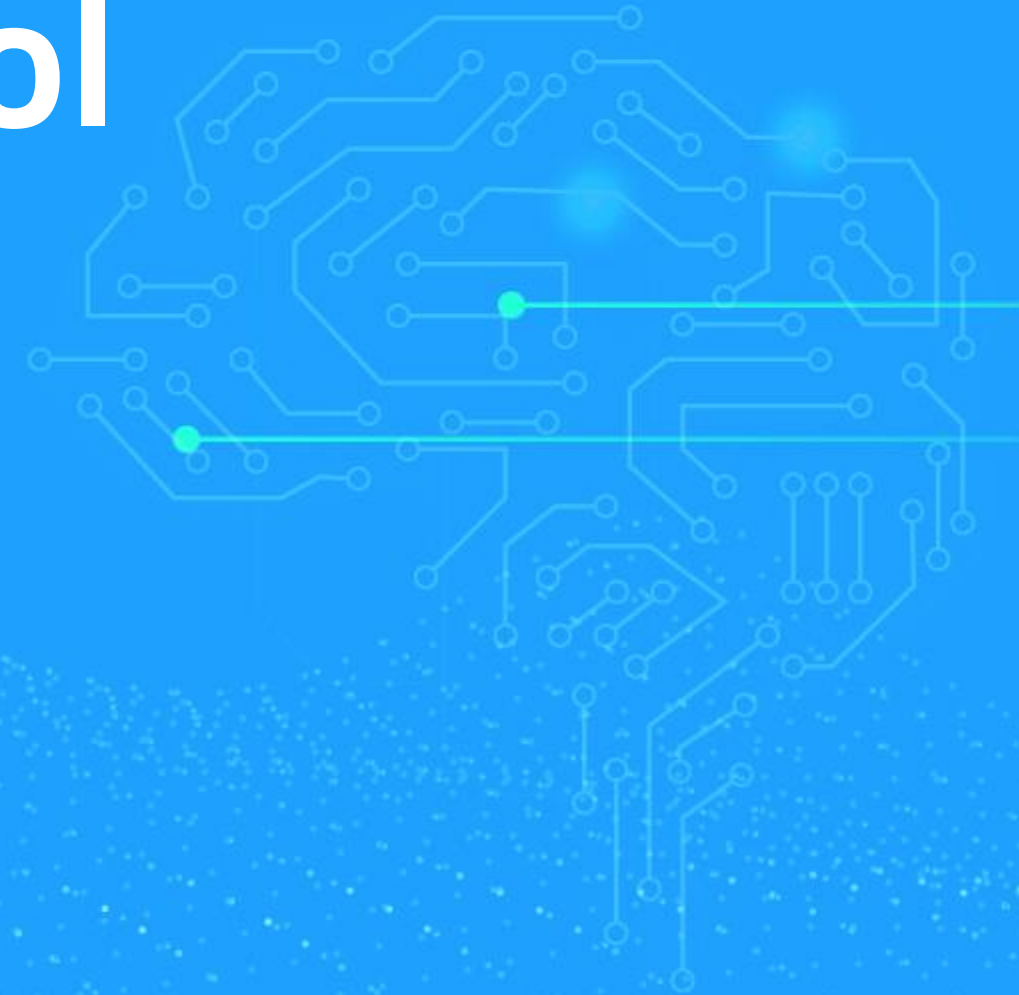
MIND



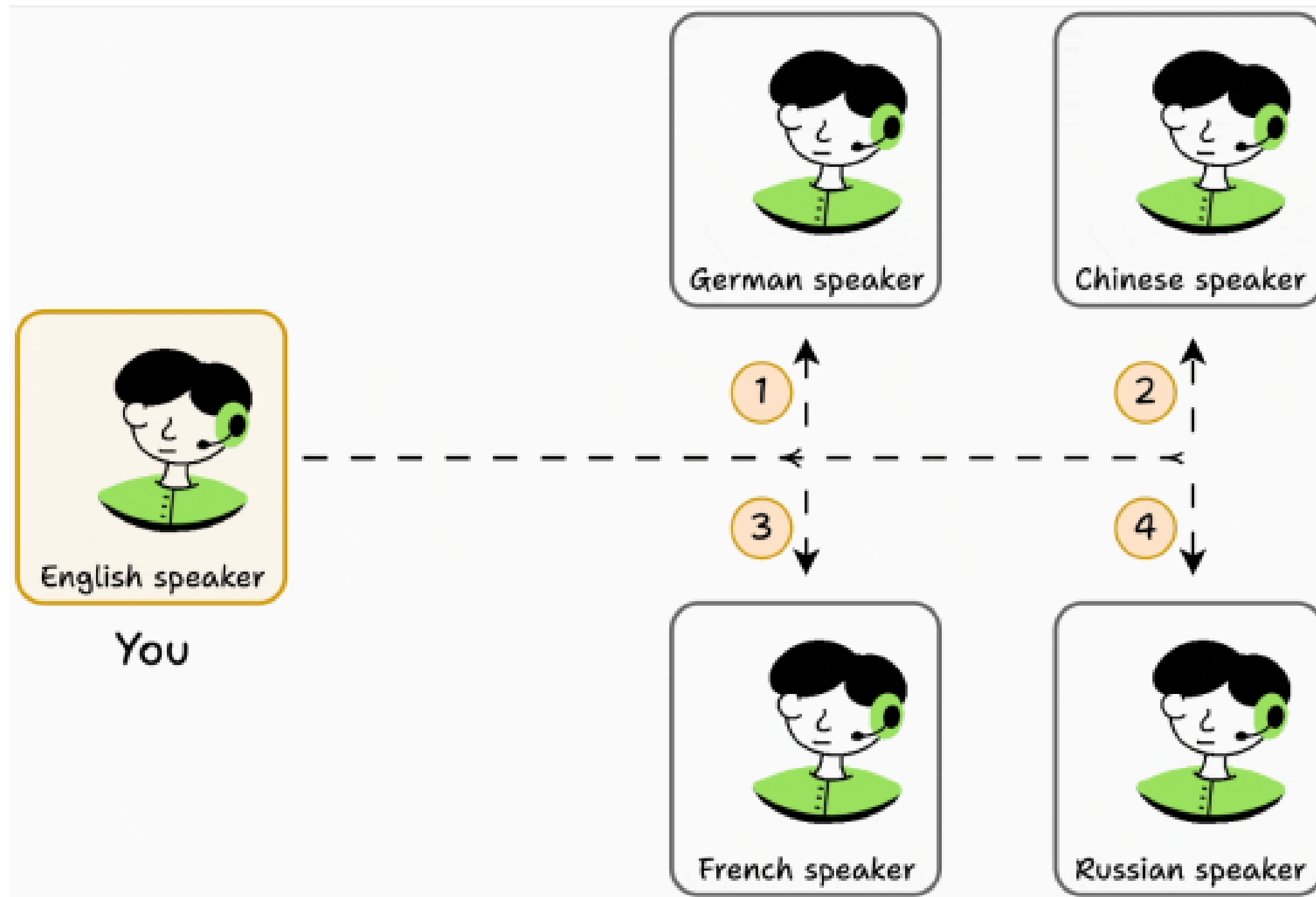
The mind encompasses the agent's **knowledge**, integrating past experiences and learned information to improve future decision-making processes.



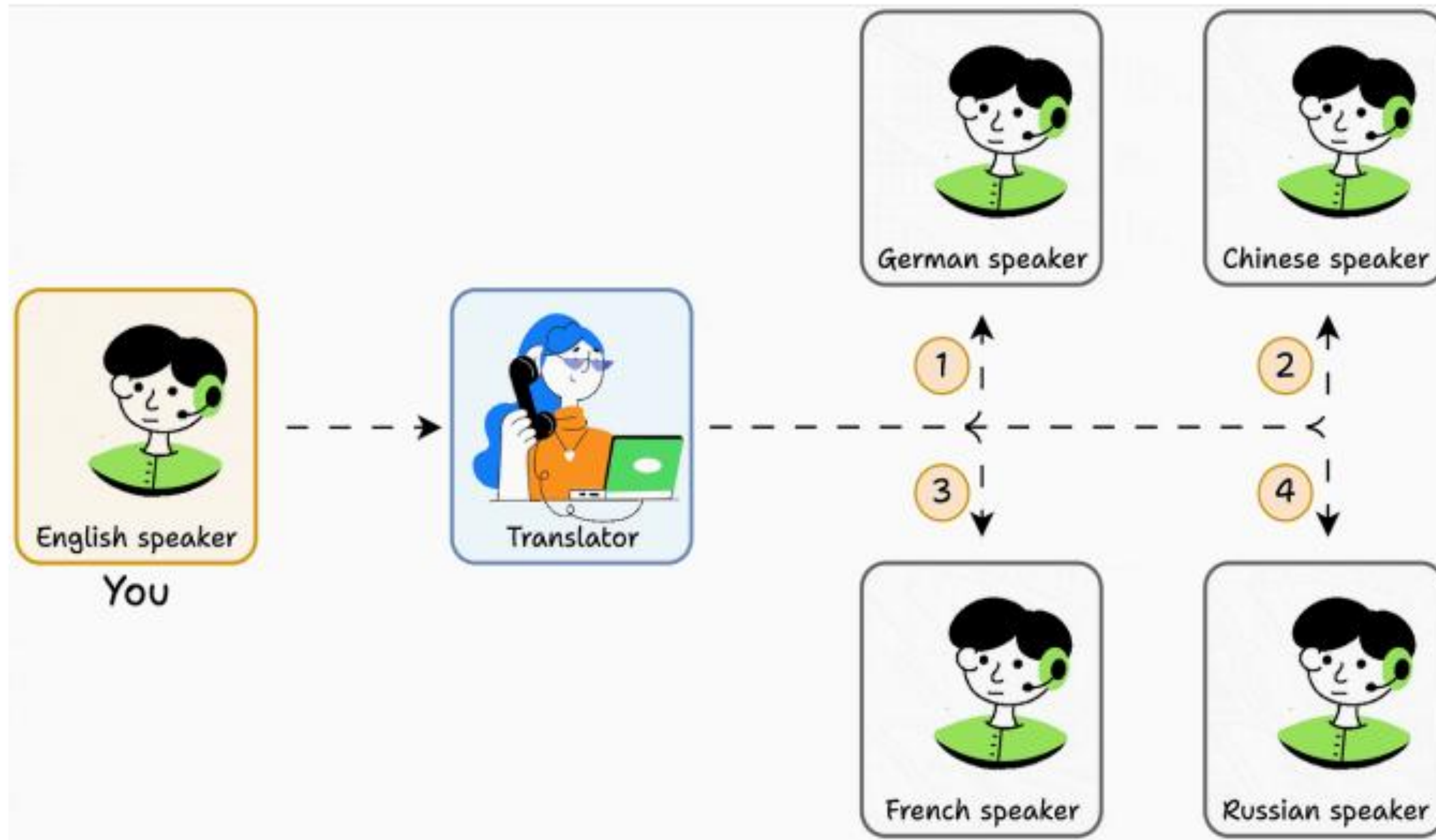
Model Context Protocol



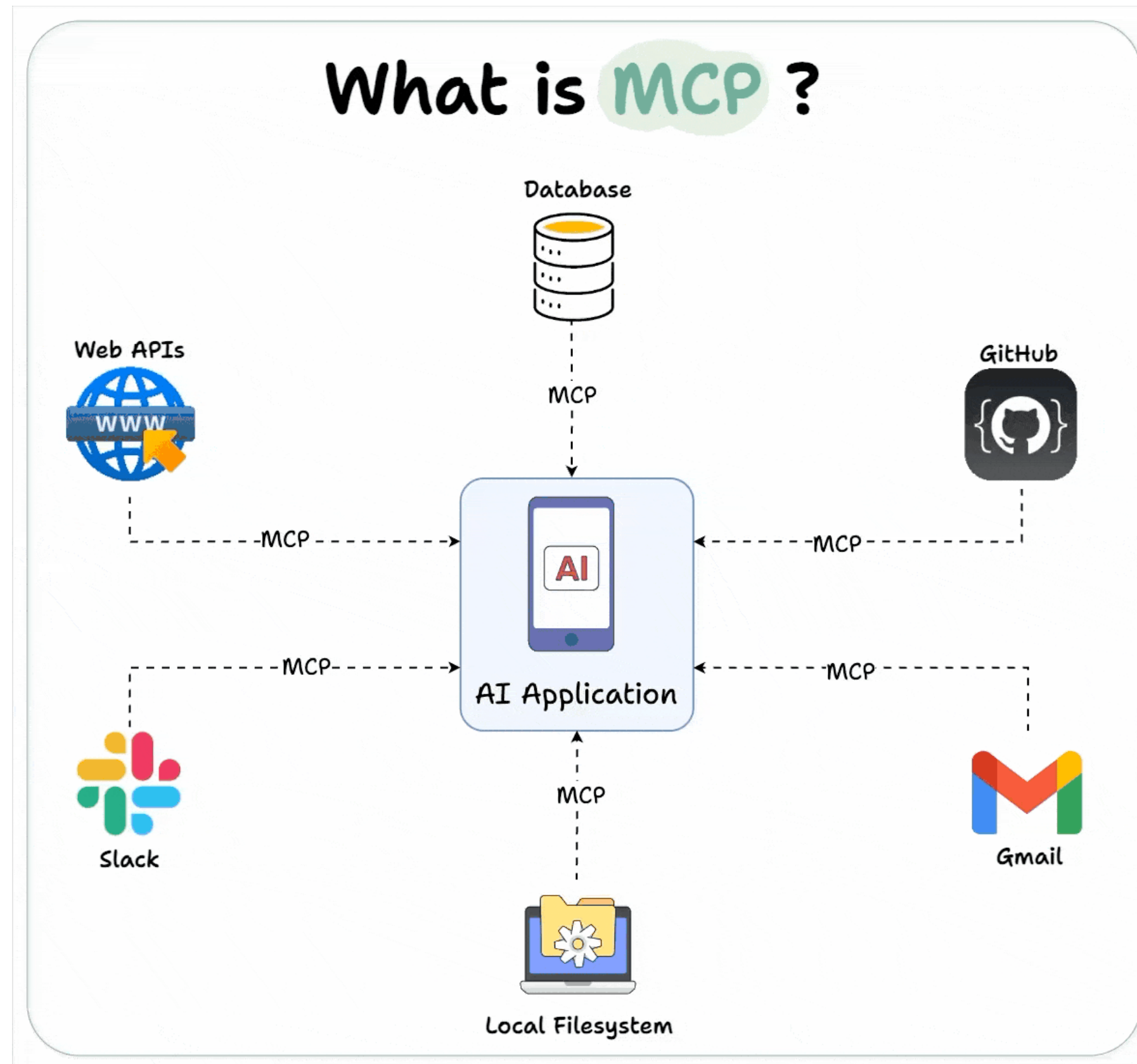
What is MCP?



What is MCP?

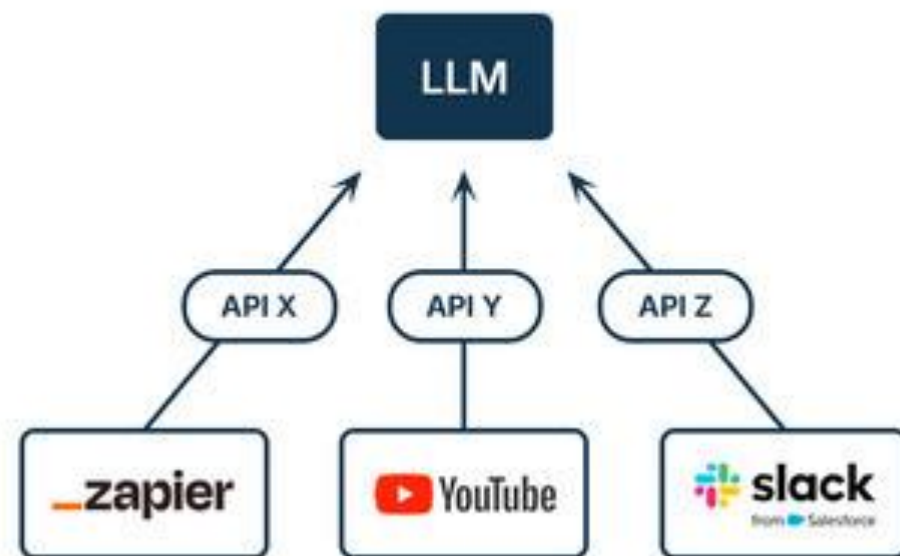


What is MCP?

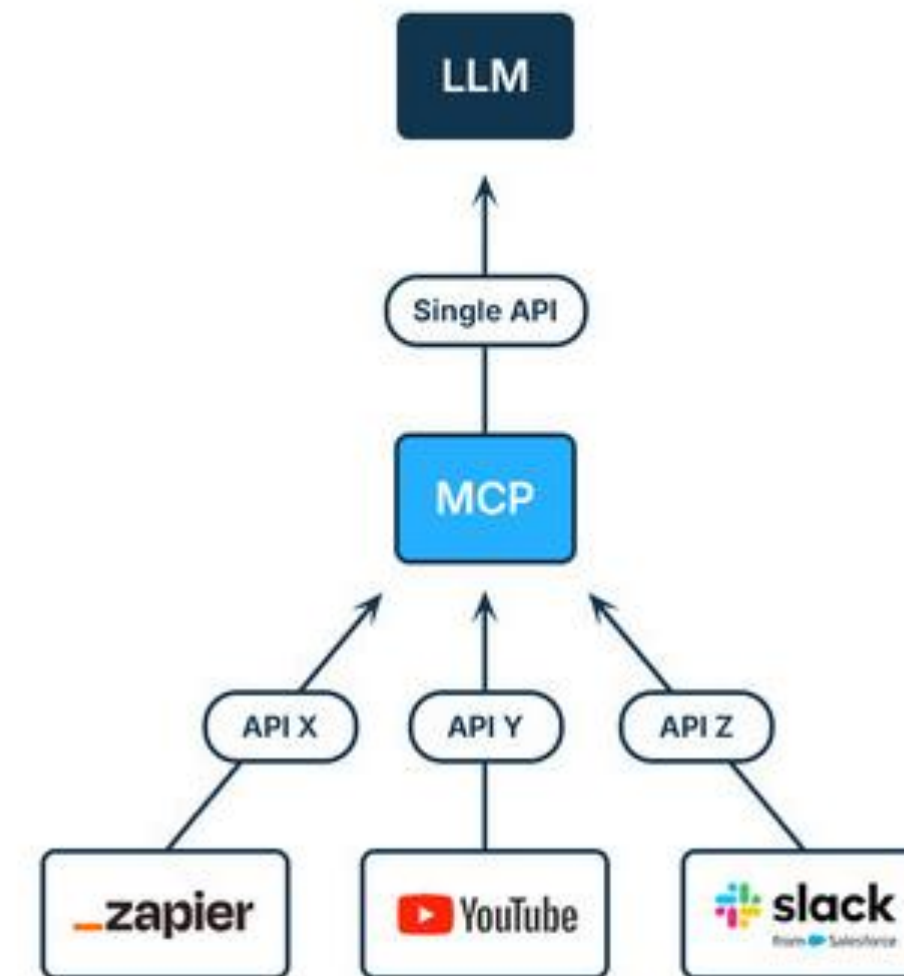


Why was MCP created?

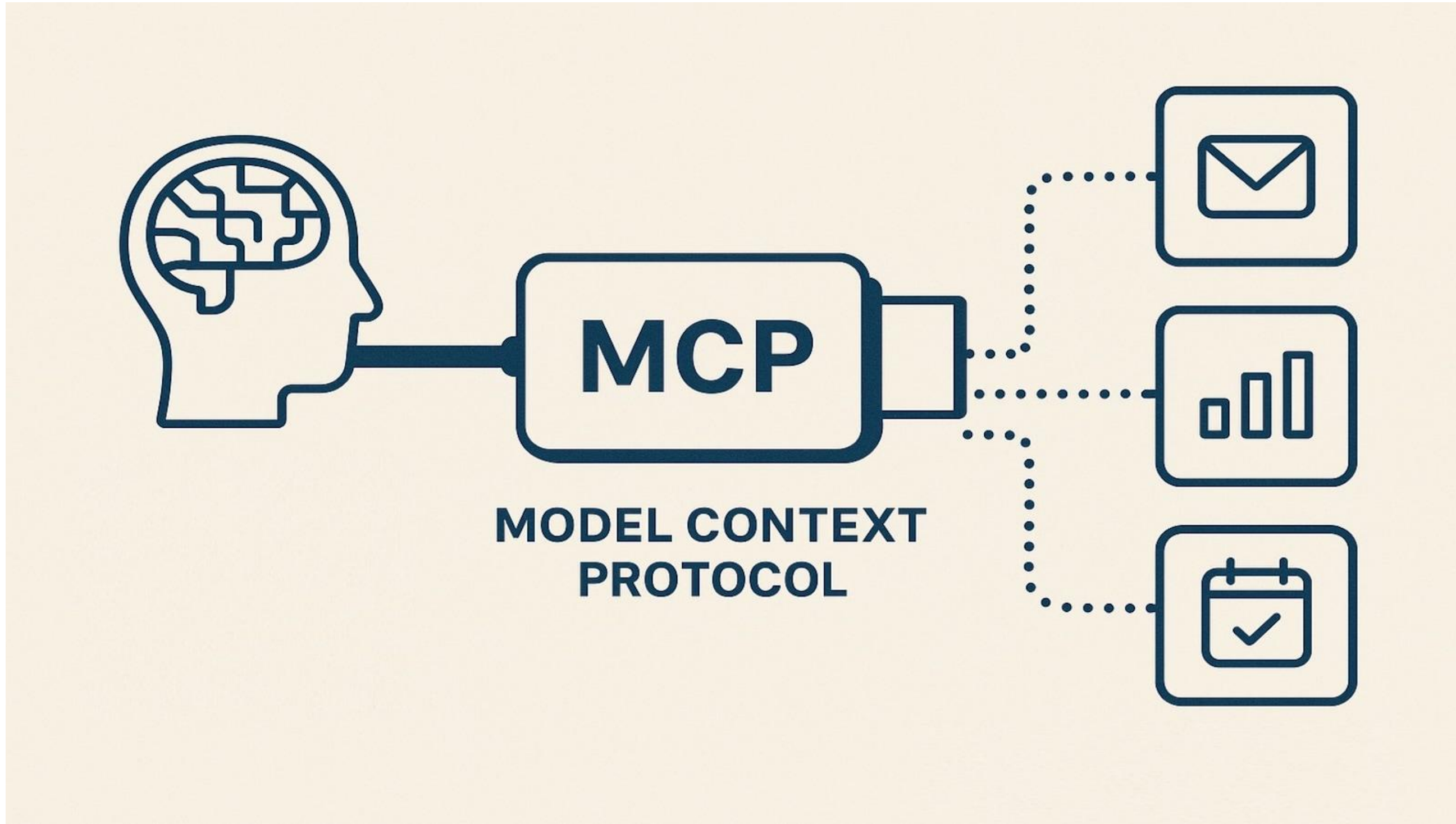
Before MCP



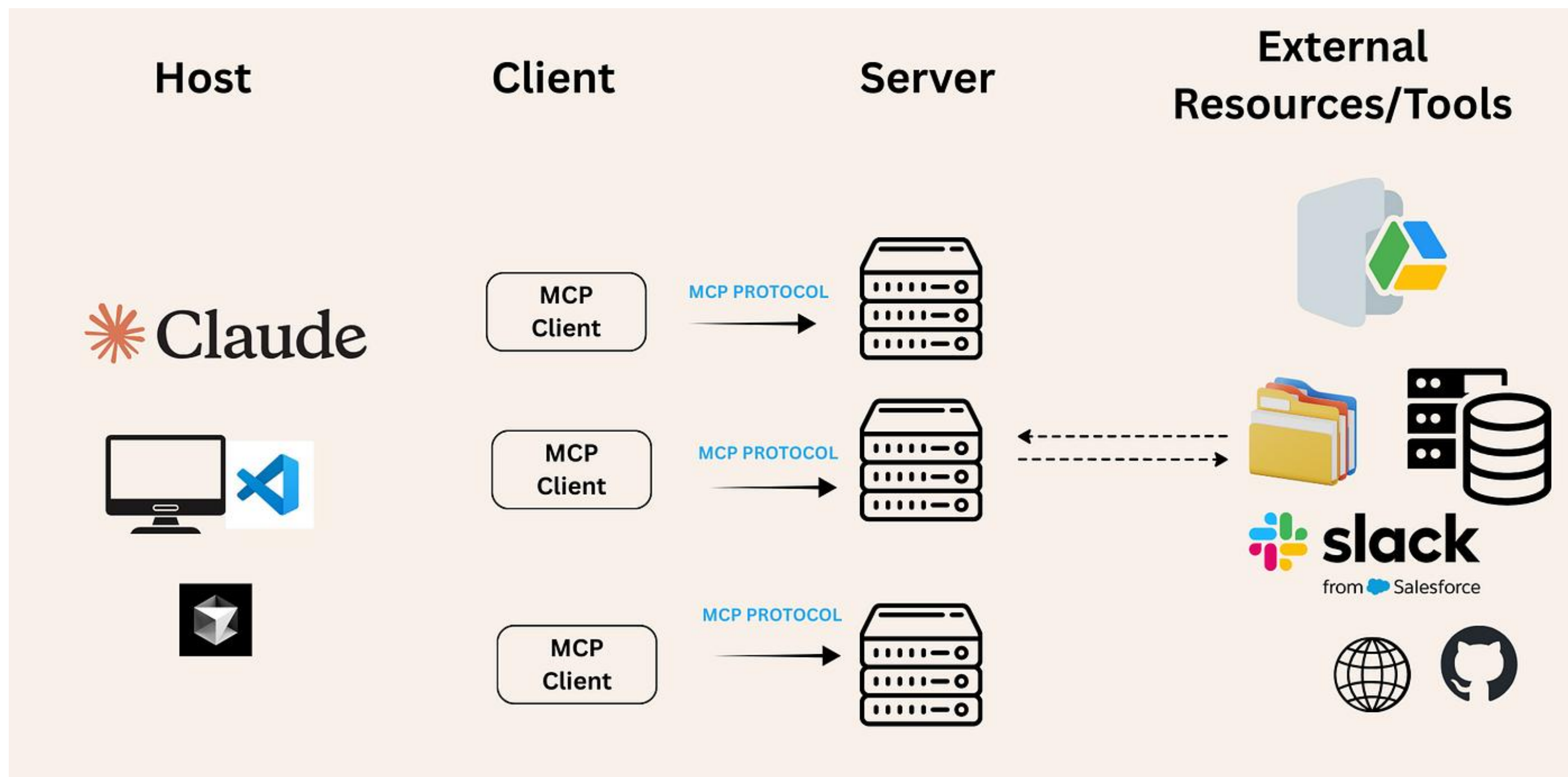
After MCP



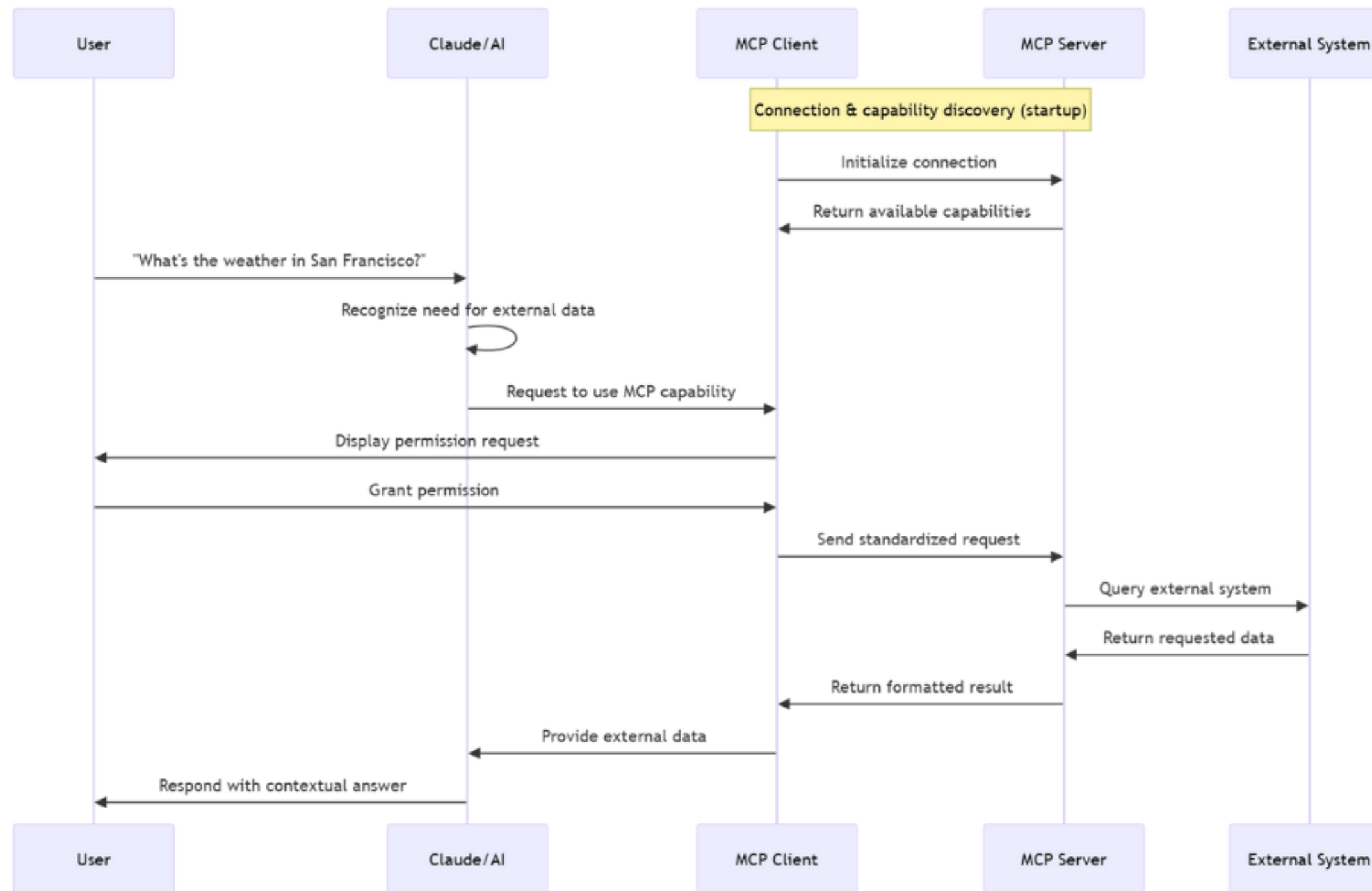
Why was MCP created?

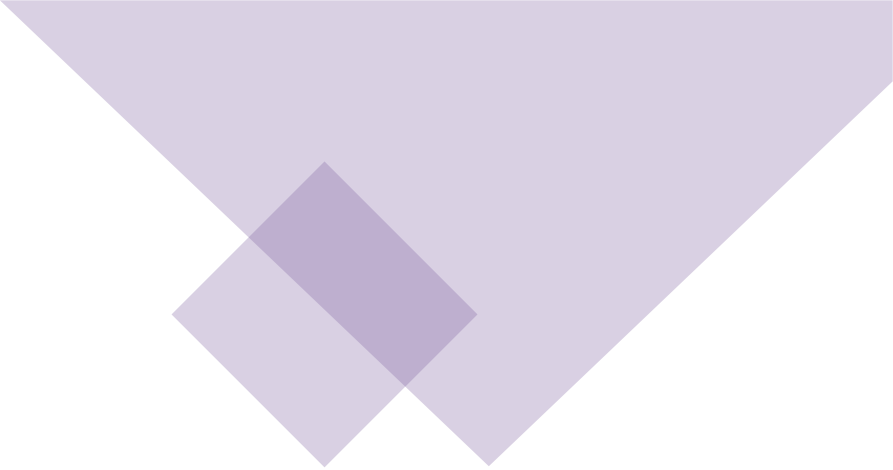


MCP Architecture

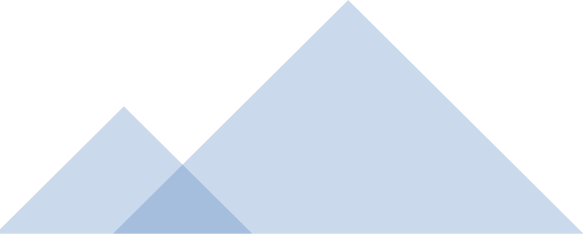


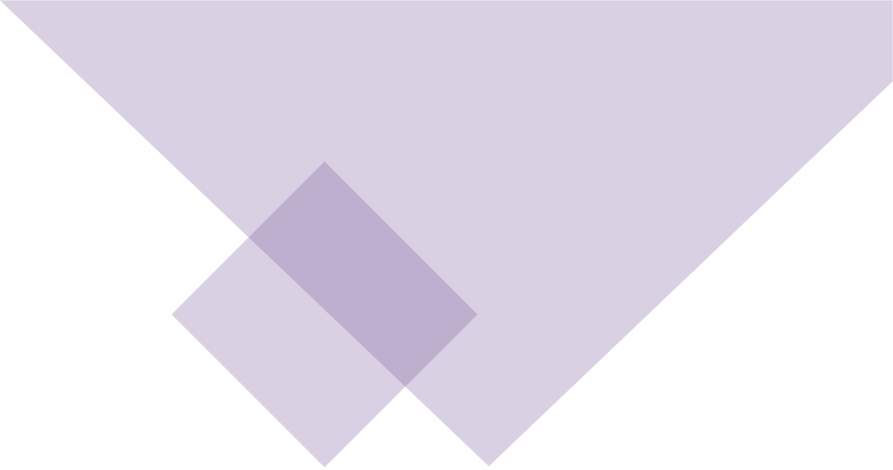
MCP Workflow



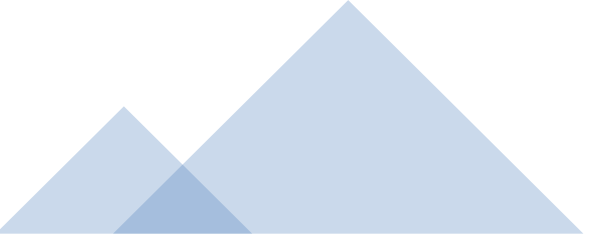


Demo



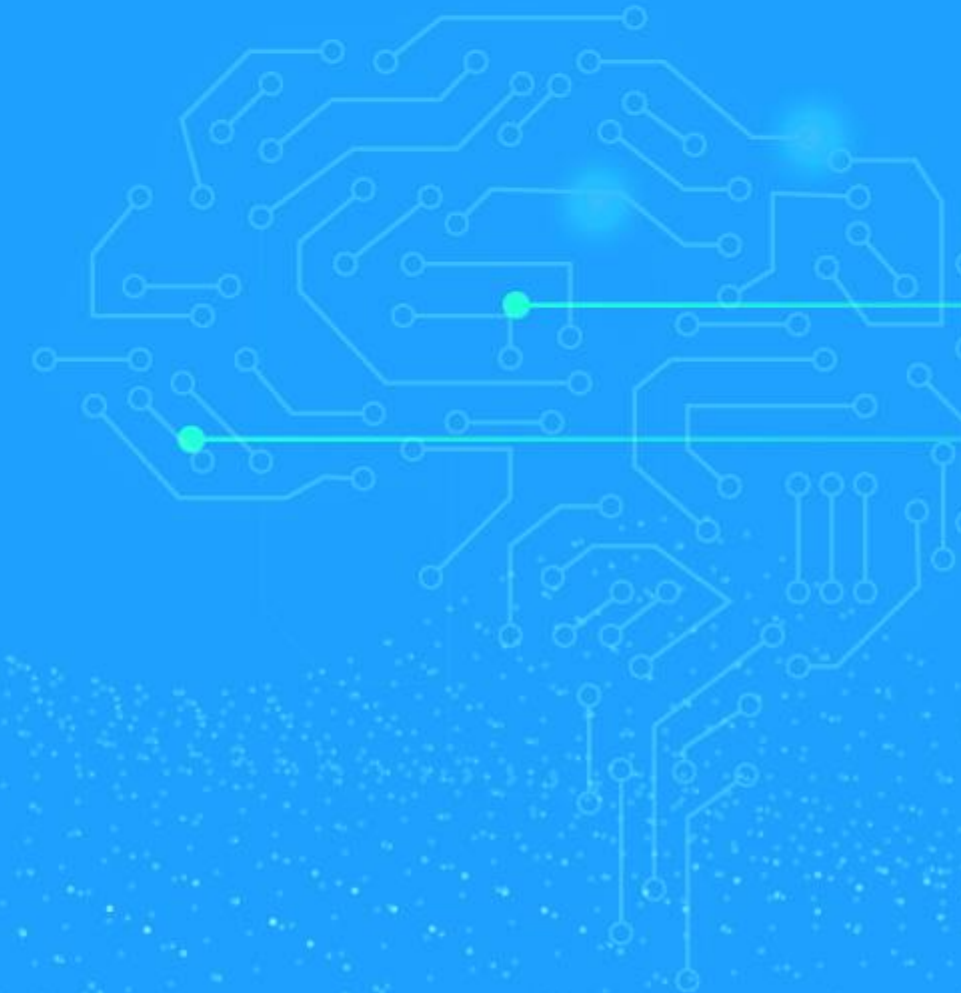


Questions & Answers





Guardrails

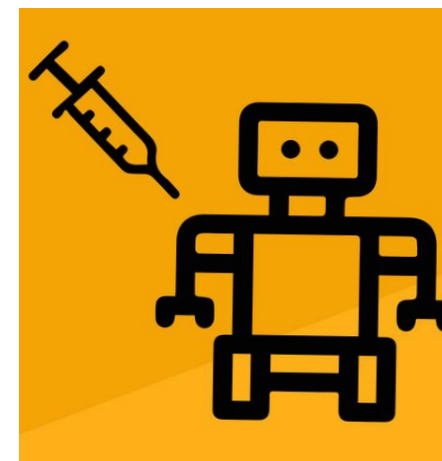


Problem statement



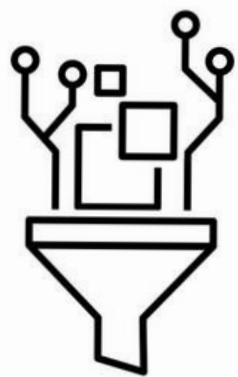
Hallucination & Ungrounded Responses

LLMs may generate fluent but factually incorrect or unverified responses, especially when lacking reliable grounding sources.



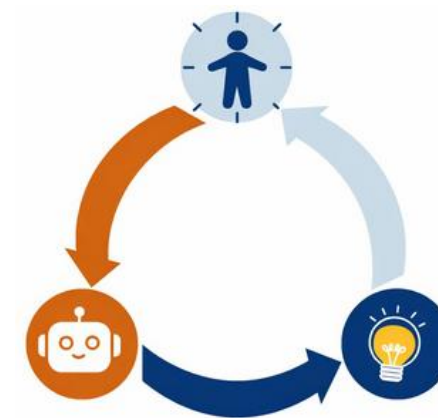
Prompt Injection & Jailbreak Attacks

Malicious inputs can manipulate system instructions, causing LLMs to bypass safety rules or perform unintended actions.



Uncontrolled or Non-Compliant Outputs

LLM outputs may violate policy, format, or regulatory constraints without explicit enforcement mechanisms.



Risky Autonomous Actions

Autonomous LLM-driven actions can lead to irreversible or high-impact outcomes without appropriate human oversight.

Solution: LLM Guardrails



A “guardrail” is a secondary “check” or validation around the input or output of an LLM model



The validation ensures that the behavior of the LLM call is valid where validity changes under the context of the application



Validity could mean no hallucinations, no PII leakage, robustness to jailbreaking, etc

What are Guardrails?

Guardrails in AI (LLM Applications) are the safety boundaries

protective limits and control mechanisms—that keep a model's behavior on the right, safe, and controlled path.

Guardrails are rules + safety nets

ensure LLMs stay safe, correct, consistent, and under your control just like a seatbelt keeps you safe in a fast car

They ensure:

Correctness (data types, structure)

Safety (no toxic, offensive, PII leakage)

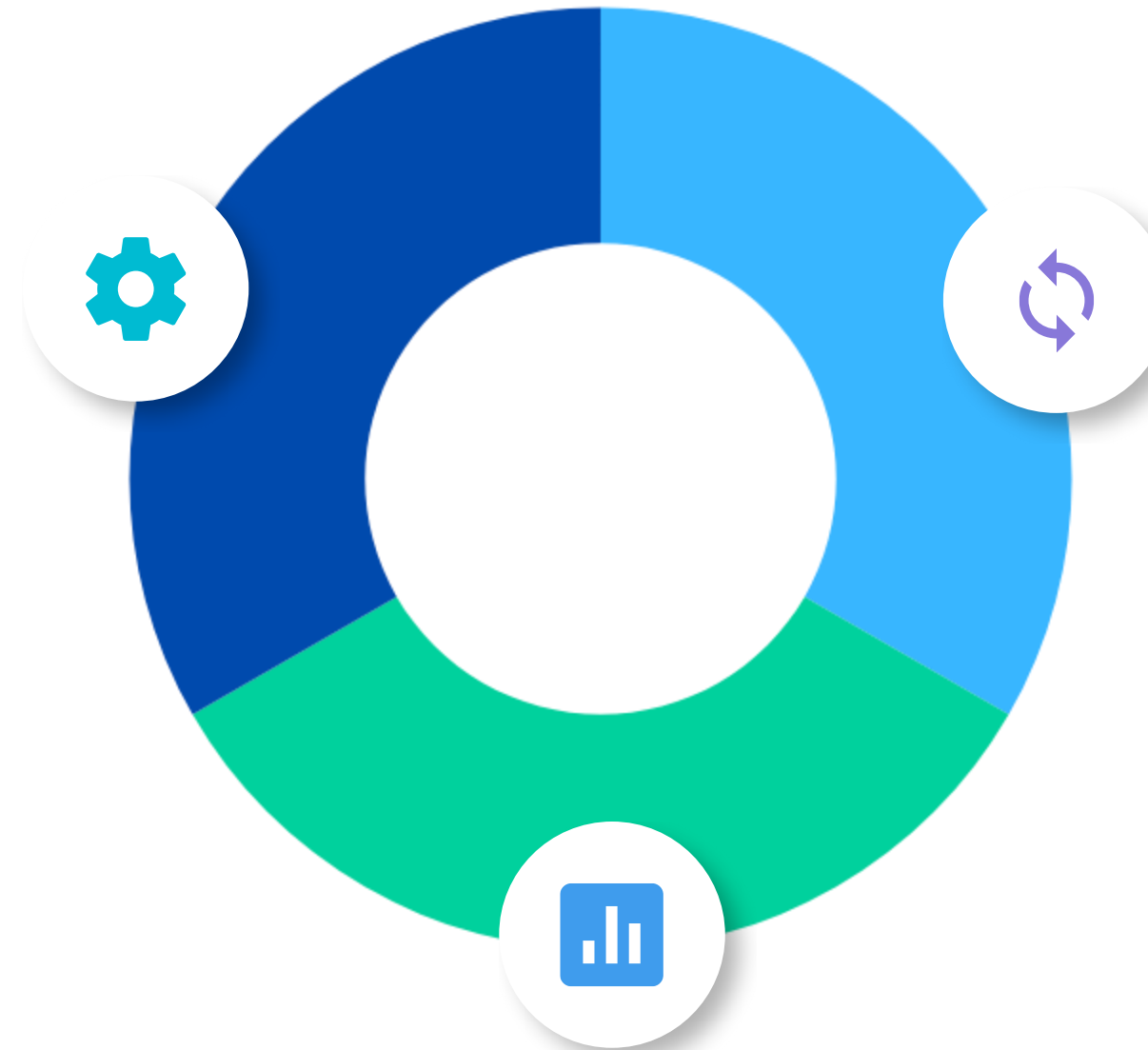
Controllability (you define the boundaries, LLM follows)

Consistency (output always in expected format)

Benefits of Guardrails

Build more complex AI workflows (e.g. agents)

- Each intermediate step is constrained and reliable, so multi-step applications can be enabled
- E.g. Step 1 guardrails, Step 2 guardrails,, Step n guardrails



Limit worse case behavior

Explicitly verify that the inviolable constraints for your application aren't violated, e.g. don't leak PII

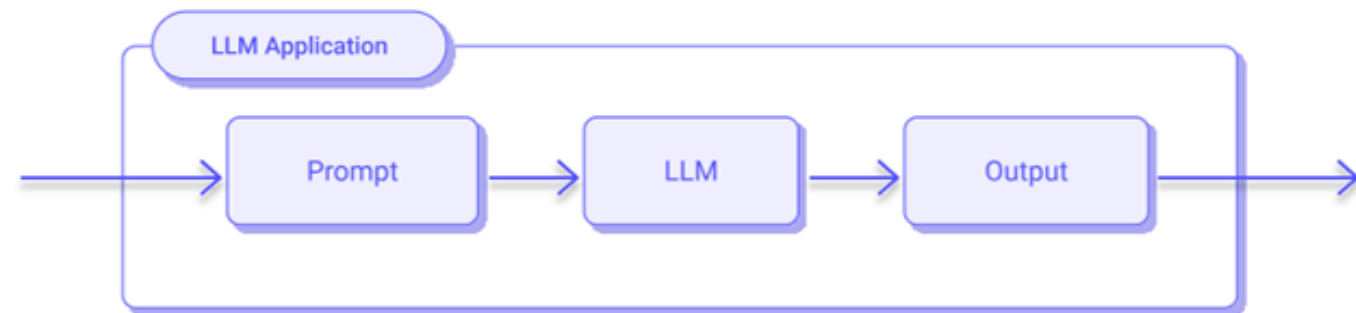
Measure the occurrence of undesirable behavior

Measure LLM refusals, log guardrails violations

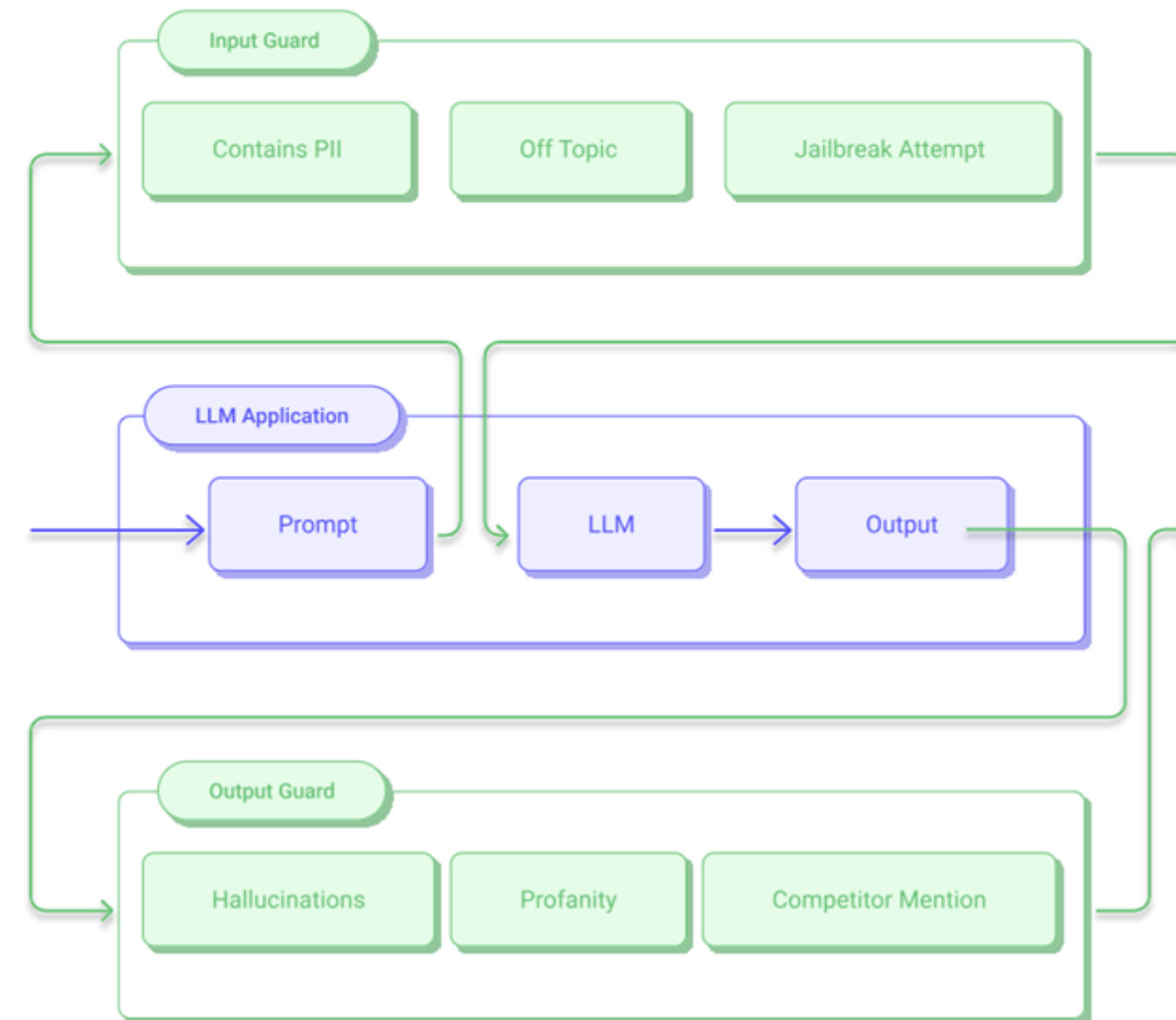


Where to apply Guardrails

Without Guardrails

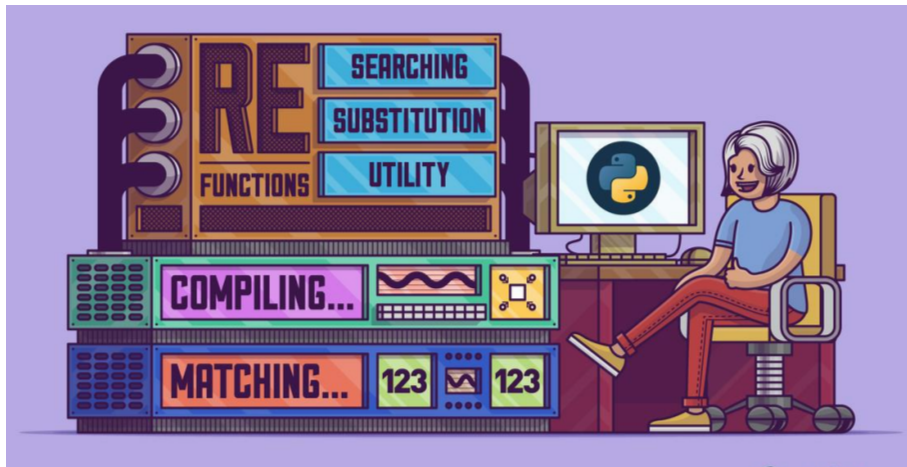


With Guardrails



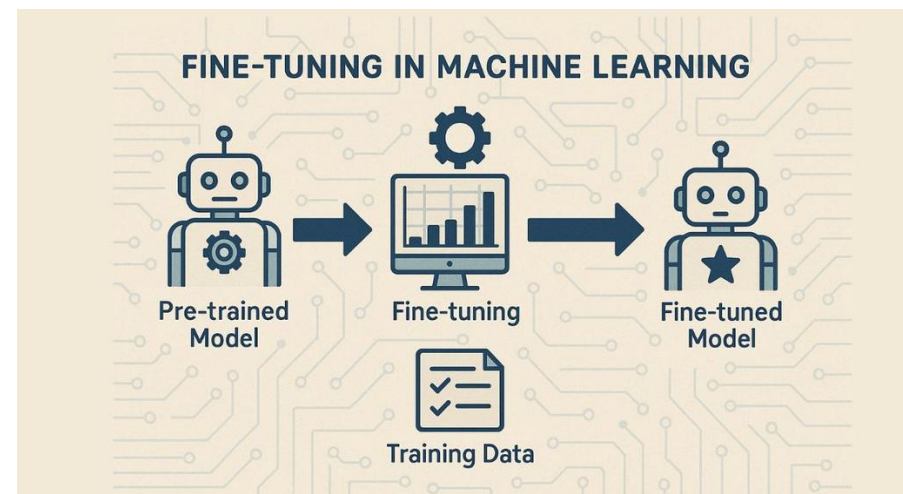
How to apply Guardrails

Approaches



RULES OR HEURISTIC SYSTEMS

Regular Expressions
Pattern Matching
Keywords / Filters



SMALL FINETUNED ML MODELS

Classification
Factuality
Topic Detection
Named Entity Recognition



SECONDARY LLM CALL

Score for Toxicity
Rate tone of voice
Verify coherence

How to apply Guardrails

Libraries and frameworks



GUARDRAILS-AI

- A Python framework that helps build reliable AI applications
- Link: <https://www.guardrailsai.com/docs>



NEMO GUARDRAILS

- Enables developers to add programmable guardrails between application code and LLMs
- Link: <https://docs.nvidia.com/nemo/guardrails/latest/index.html>



OPENAI GUARDRAILS

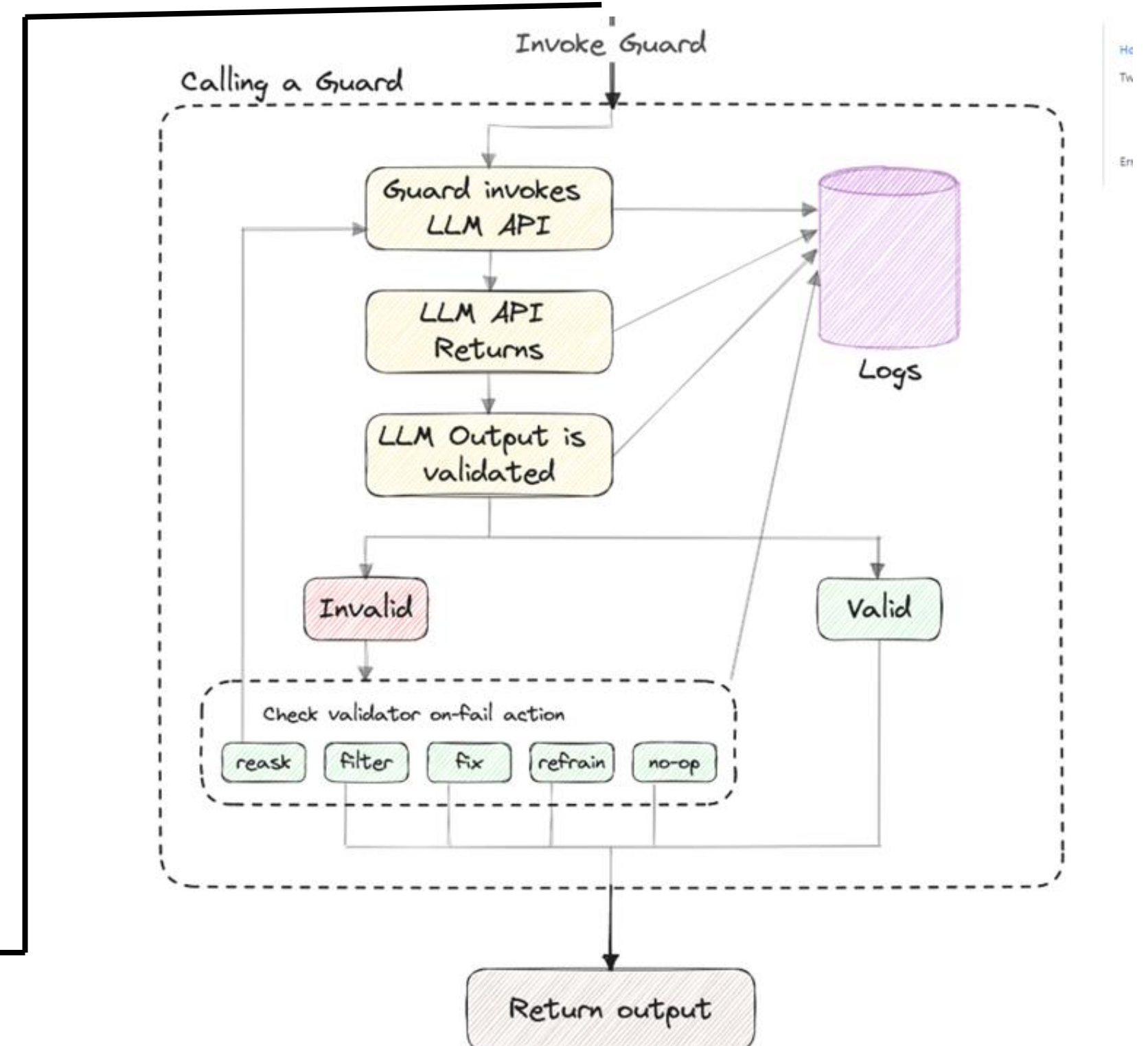
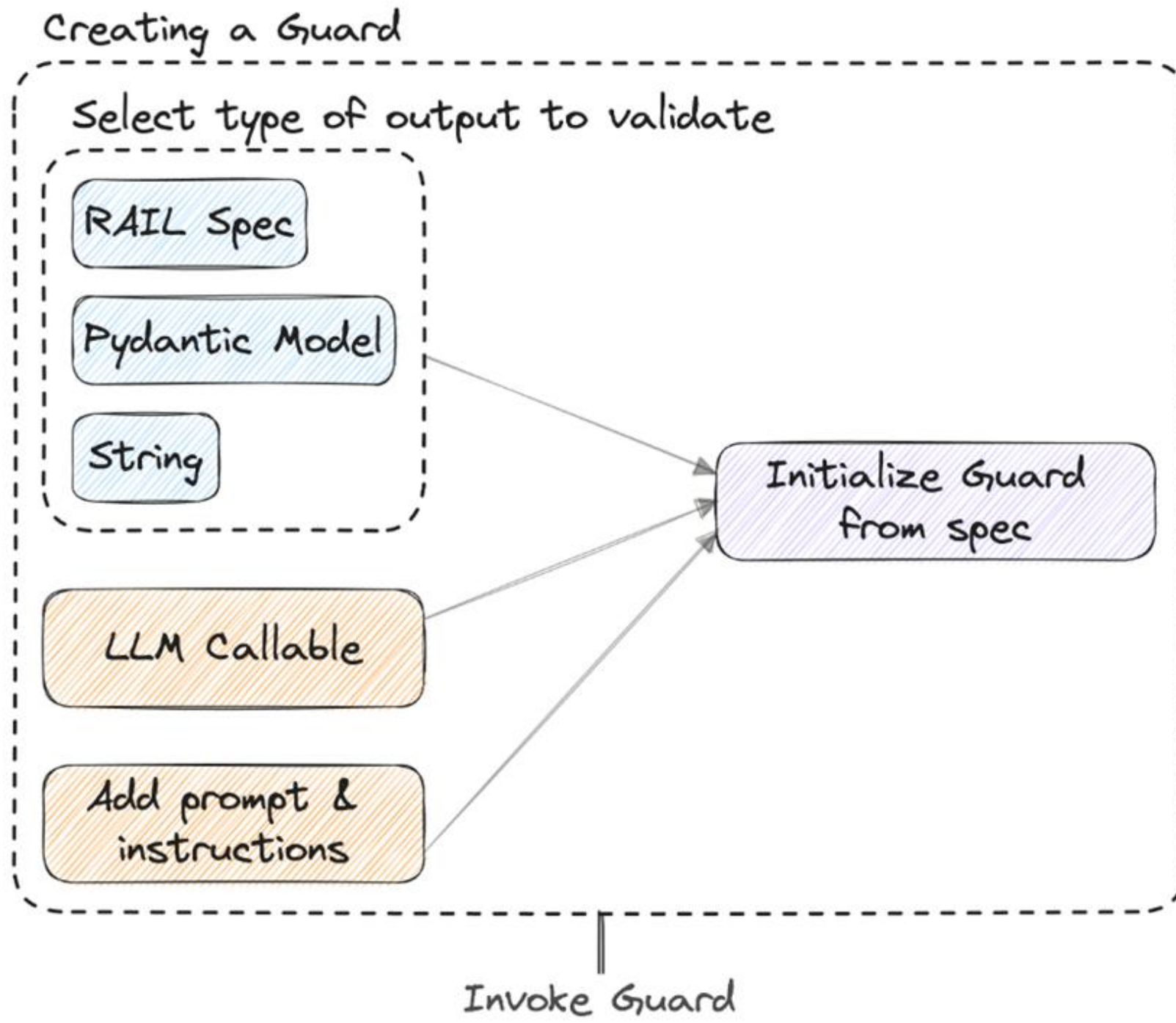
- A safety framework for LLM applications that automatically validates inputs and outputs using configurable checks
- Link: <https://openai.github.io/openai-guardrails-python/>



LMQL (LANGUAGE MODEL QUERY LANGUAGE)

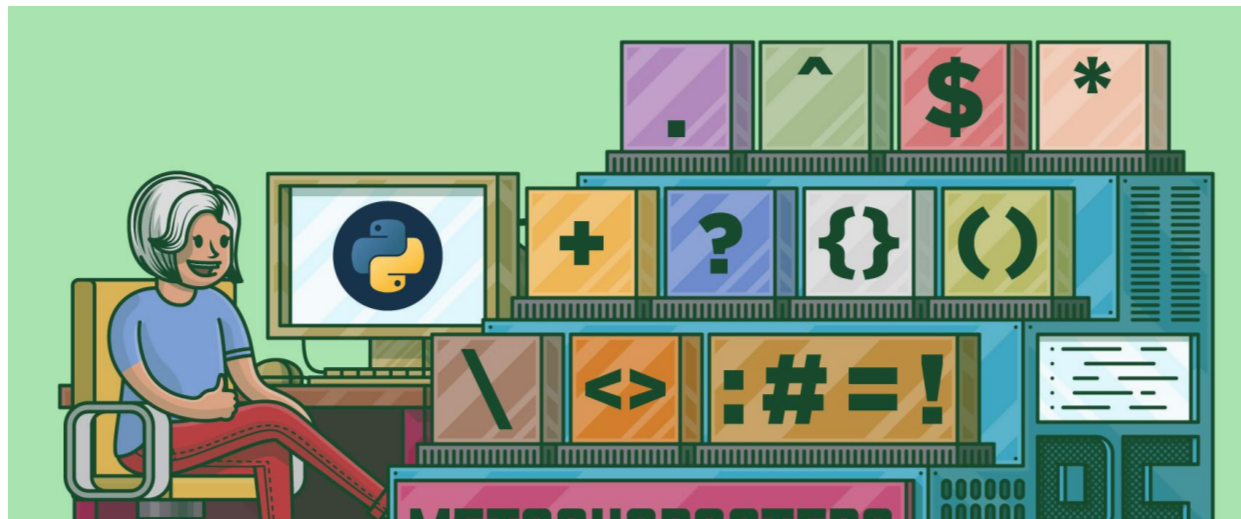
- Robust and modular LLM prompting using types, templates, constraints and an optimizing runtime.
- Link: <https://lmql.ai/docs/>

How – Guardrails AI



How – Guardrails AI

Guardrails can be implemented using two complementary approaches



DETERMINISTIC GUARDRAILS

- Rule-based logic:
 - Regex patterns
 - Keyword matching
 - Explicit checks.
- Fast, predictable, and cost-effective
- May miss nuanced violations.



MODEL-BASED GUARDRAILS

- Use LLMs or classifiers to evaluate content with semantic understanding.
- Catch subtle issues that rules miss
- Slower and more expensive.

How - Built-in guardrails

PII detection - Built-in PII types

mac_address - MAC addresses



url - URLs



email - Email addresses



ip - IP addresses



credit_card - Credit card
numbers (Luhn validated)

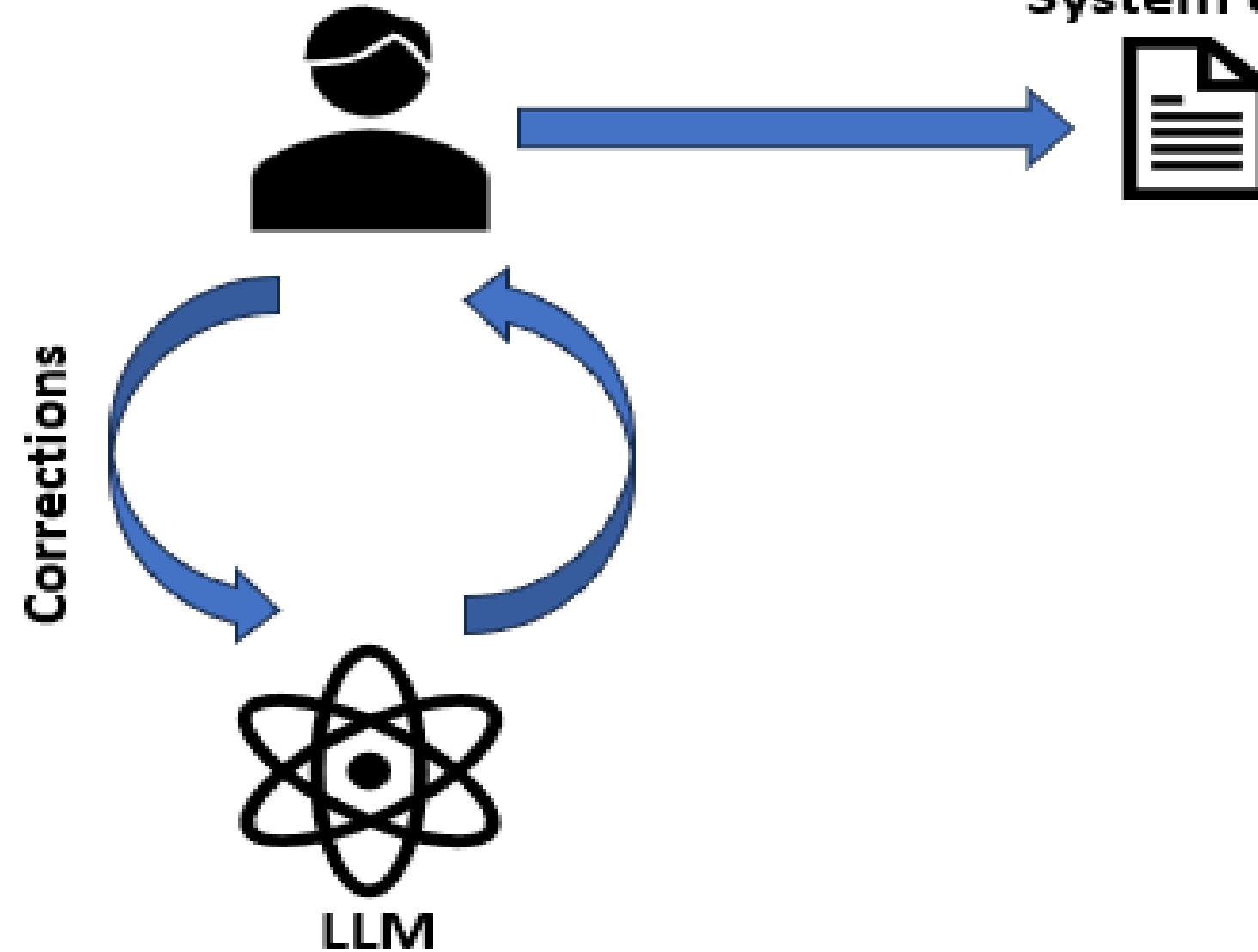


How - Built-in guardrails

Human-in-the-loop

Output review by Human

System output



How - Custom guardrails

BEFORE AGENT GUARDRAILS

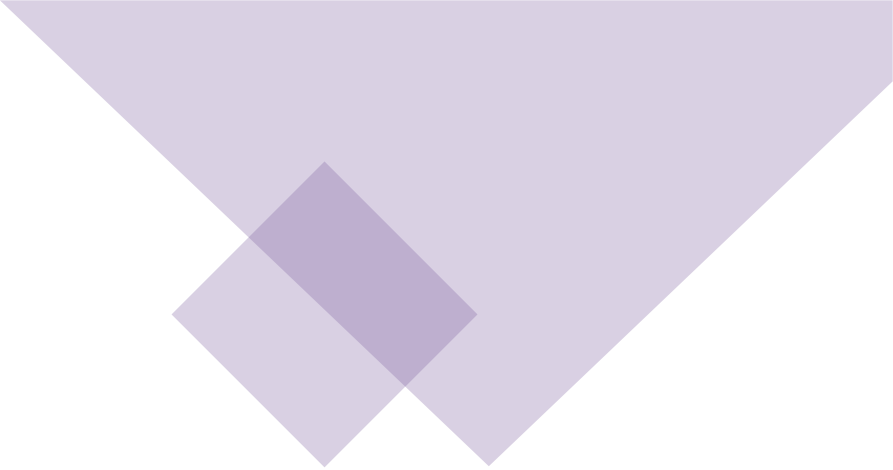
- ❑ Use “before agent” hooks to validate requests once at the start of each invocation.
- ❑ Useful for session-level checks: authentication, rate limiting, or blocking inappropriate requests before any processing begins.

AFTER AGENT GUARDRAILS

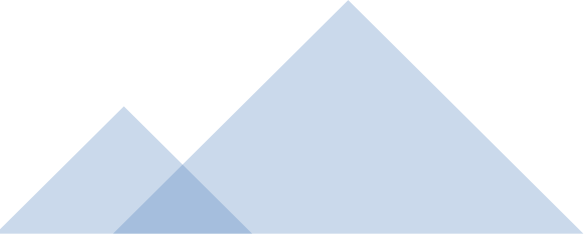
- ❑ Use “after agent” hooks to validate final outputs once before returning to the user.
- ❑ Useful for model-based safety checks, quality validation, or final compliance scans on the complete agent response.

COMBINE MULTIPLE GUARDRAILS

- ❑ Stack multiple guardrails by adding them to the middleware array.
- ❑ They execute in order, allowing you to build layered protection.



Demo



Guardrails on Cloud - Amazon Bedrock

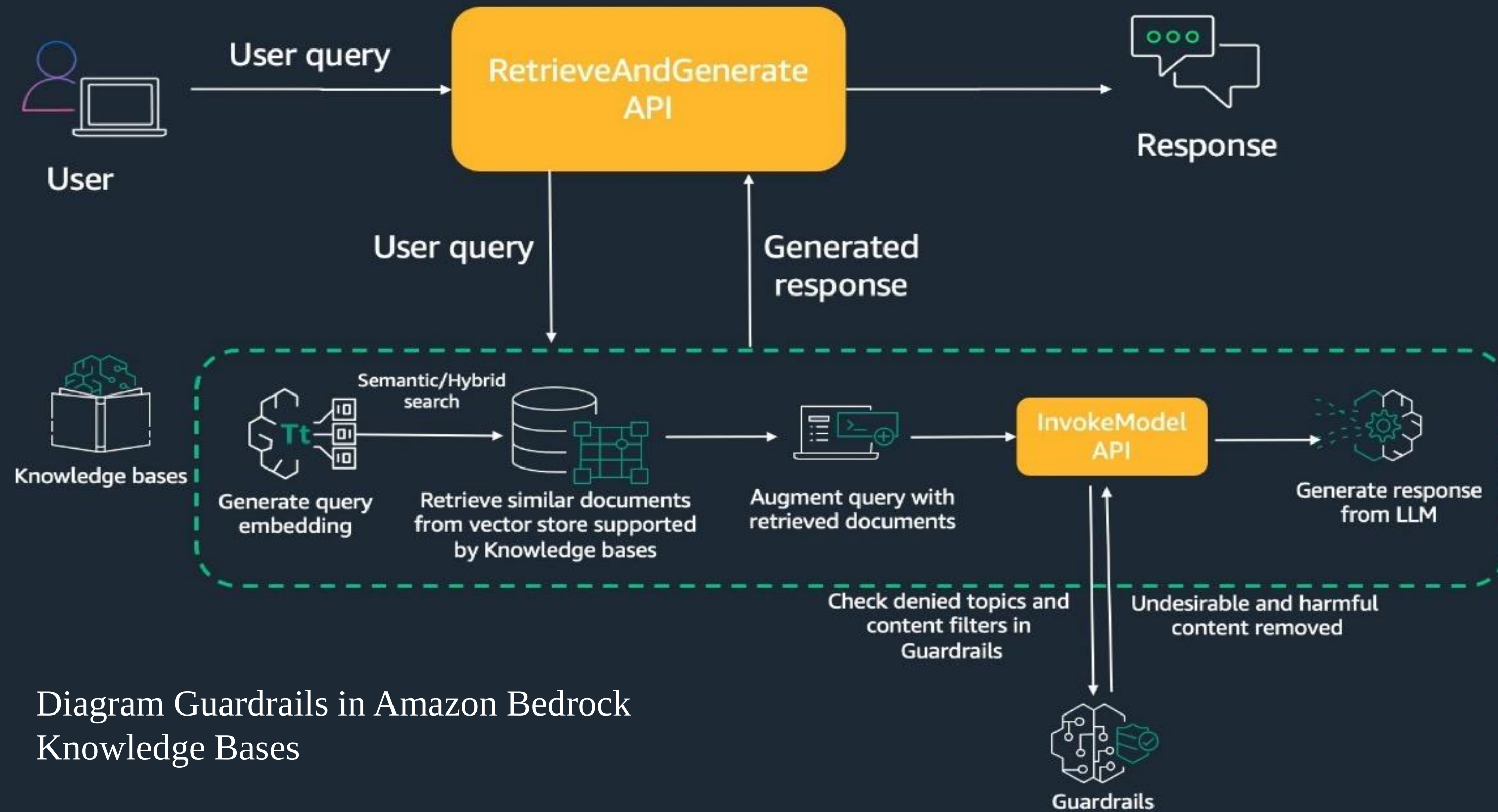
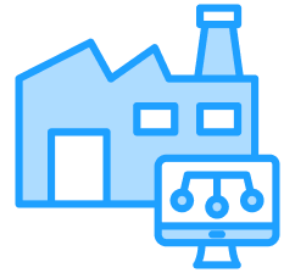


Diagram Guardrails in Amazon Bedrock Knowledge Bases

Guardrails on Cloud - Amazon Bedrock



**Topic-based
filtering**



Keyword filtering



Profanity filtering

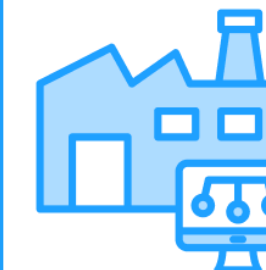
- Content Filtering for both user request and response
- Supported Models: Text and Image model
- Integration: Attach directly to Bedrock Agents or Knowledge Bases



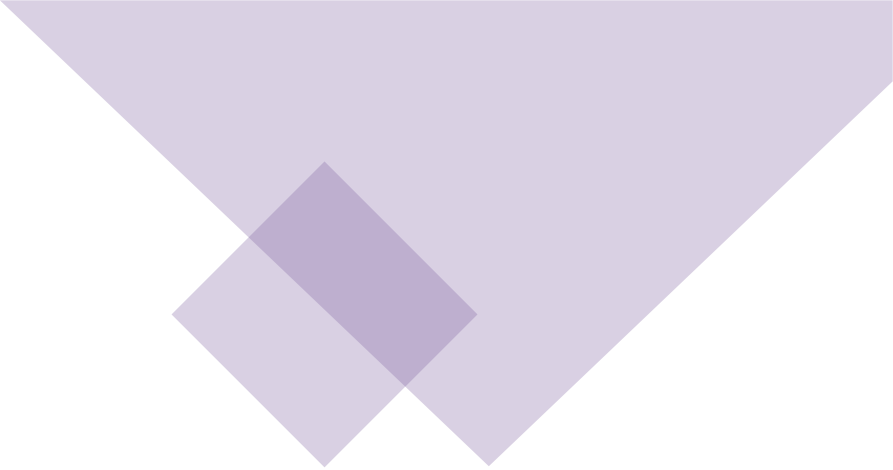
**Contextual
Grounding Check:
reduce
hallucinations**



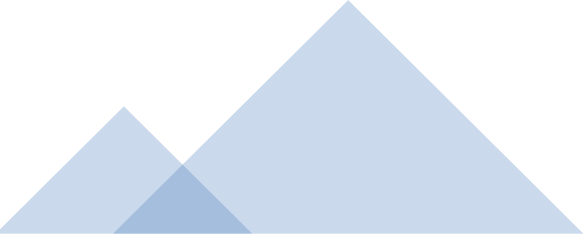
**Remove or mask
PII (Personally
Identifiable
Information)**

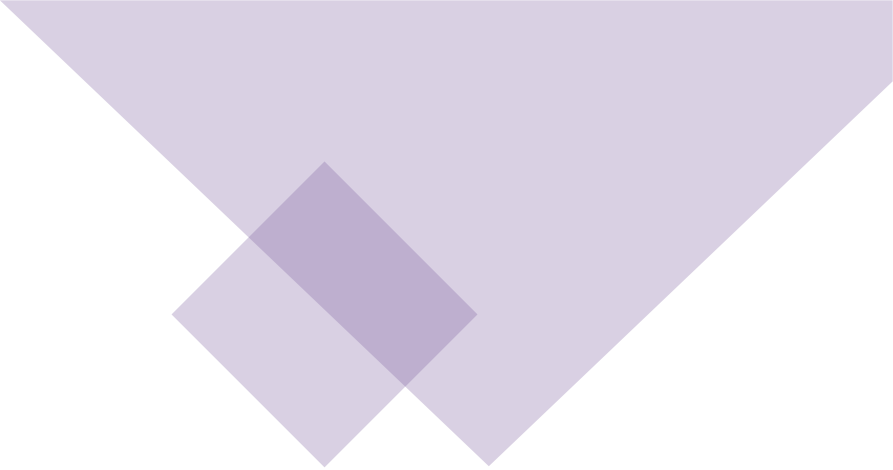


**Harmful
categories
filtering**

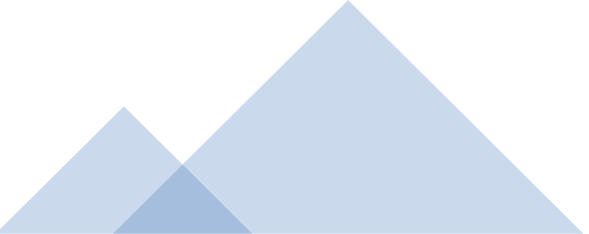


Demo





Questions & Answers





THANK YOU!

Vietnam: **+84 28 3997 8000**

North America: **+1 844 224 4188**

Europe: **+49 171 617 3356**

Email: **sales@tmasolutions.com**

Australia: **+61 478 407 408**

Japan: **+81 364 324 994**

Website: **www.tmasolutions.com**